

Development guidelines applicable for
TC501/701 TC201 ET401

Connection Modes

- Two connection modes are supported.
 - RFID_ONLY mode
 - In RFID only mode cellular stack will not be available until RFID only mode is disconnected
 - Will not be able to receive any data/cellular call
 - RFID+WAN mode
 - In this mode data can co-exist with RFID operation.
 - User can receive cellular call. During cellular call RFID operation will not be available
- Default connection API will initiate RFID_ONLY mode
- Additional connect API provided to initiate connection in RFID+ WAN mode

```
void connect(ENUM_CONNECTION_MODE mode)
```

WWAN + RFID use cases: SIM combination

SIM1	SIM2(eSim)	RFID and WAN co-existence behavior
Configured	Not Configured	RFID and cellular call cannot co-exist. Can receive VOIP call
Not configured	Configured	RFID and cellular call cannot co-exist. Can receive VOIP call
Configured with data preference	Configured	RFID and Data apps or VOIP app can co -exist. RFID and cellular call cannot co-exist When RFID is active SIM2 will be disabled
Configured	Configured with data preference	RFID and Data apps or VOIP app can co -exist. RFID and cellular call cannot co-exist When RFID is active SIM1 will be disabled

WWAN + RFID use cases : Connection modes

RFID ONLY mode	RFID+WWAN mode
WIFI active: RFID + Data transmit -> both working	WIFI active: RFID + Data transmit -> both working
WIFI active: RFID + VOIP call -> RFID stop, VOIP call received (+proximity)	WIFI active: RFID + VOIP call -> RFID stop, VOIP call received (+proximity)
LTE active: RFID + Data transmit -> RFID continue, no data transmission	LTE active: RFID + Data transmit -> both working
LTE active: RFID + phone call -> RFID continue, no phone call	LTE active: RFID + phone call -> RFID stop, phone call received
suspend/resume: RFID continue reading	suspend/resume: RFID continue reading

WWAN + RFID use cases: Read rate impact

WAN State	Impact on RFID
Idle	✓ Best performance — full RFID throughput
Hold Voice/Data	⚠ Reduced — ~50% <u>drop in</u> read rate
Active Voice	✗ RFID not operational

Conditions when RFID connections will fail

- USB cable plugged
- Low battery below 10%
- Airplane mode
- Will automatically disconnect if idle for more than 9 mins
- Will automatically disconnect when user receives VOIP call and near body

Multi client support

- Does not support multi client.
- If second client attempts to connect while first client is connected. Client connection will fail
- It's the same reason you cannot have both Data wedge and 123RFID app connected at the same time

Features not Supported

- AB flip configuration is not supported
- No Firmware update
- Periodic start/stop trigger not supported
- CSP support is limited to factory reset and region configuration
- Can get operation failure exceptions if we try to configure new regions that is supported from older 2.0.5.275 SDK
- Data wedge by default always connects in RFID+WAN mode

Recommended settings by use cases

- inventory operations in small tags population
[session: S0 or S1, link: PR-ASK M4 40KHz or PR-ASK M2 80KHz, Tx power 25 dBm]
- inventory operations in large tags population
[session: S2, S3, link: PR-ASK M1 160KHz or PR-ASK M2 80KHz, Tx power 25 dBm]
- read/write operations on specific tags
[session: S0, link: PR-ASK M4 40KHz or PR-ASK M4 40KHz, reduce Tx power]